Controllable Generative Knowledge-Driven Few-Shot Object Detection From Optical Remote Sensing Imagery

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***Abstract*— Few-shot object detection (FSOD) has to learn classification and localization information for unseen object detection under very low-data resource regimes. However, when deficient samples are adopted for model training, it is hard to build powerful location-aware and identification abilities for well coping with agnostic bias from diverse testing scenarios; at the same time, the overfitting phenomenon is easily occurring. Therefore, in this article, a controllable generative knowledge-driven FSOD called CGK-FSOD is proposed for unseen object detection from optical remote sensing imagery. Specifically, to enrich the learnable data space of scarce samples for preventing incomplete agnostic-bias learning, while avoiding the overfitting phenomenon, a visual–textual prompt-based controllable data generation is designed to generate high-quality object detection data based on pretrained foundational models [i.e., the stable diffusion (SD) and contrastive language-image pre-training (CLIP)], which not only can introduce the generalized domain-level knowledge into the remote sensing domain but also sets up an all-round data space to support complete learning of potential agnostic bias. Furthermore, with respect to the denoising generative process of SD, a series of cross-modality generative features in latent representation space are reused for few-shot fine-tuning by the designed cross-modality feature embedding (CMFE), which not only can bring diverse generative abilities into the feature fusion step of the detector but also gracefully sets up feature representation scalability to make the detector better adapt to agnostic bias from diverse testing scenarios of FSOD. Finally, extensive experiments are executed on two public remote sensing datasets (e.g., DIOR and NWPUVHR-10), and the results indicate that the proposed CGK-FSOD is very effective and flexible for FSOD.**

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I. Introduction

FEW-SHOT object detection (FSOD) is a challenging task for detecting unseen novel objects, which relies only on scarce labeled training samples. Notably, it not only needs to recognize the corresponding categories that belong to the novel categories but also has to determine their precise positions from very complicated backgrounds. However, scarce training samples cannot provide adequate information to support the model training process; therefore, it is difficult to establish sufficient location-aware and identification abilities. In addition, under few-shot learning scenarios, the overfitting phenomenon easily appears at the model training step, which would lead the trained object detector to present poor generalization ability for coping with the agnostic bias from diverse testing scenarios. In practice, the FSOD ability is very important for Earth observations, such as scientific expeditions [1], endangered animal monitoring [2], and military reconnaissance [3]. Nevertheless, the detectors can only be trained on a small number of training samples from these applications, and they are also required to be well-generalized for unseen object detection. Consequently, several works [4], [5], [6], [7] have been dedicated to studying more effective few-shot learning approaches to establish a powerful FSOD ability for optical remote sensing object detection. For example, Zhao et al. [4] proposed a few-shot object detector with an involution backbone and path-aggregation feature extraction, which can support the model in learning shape bias for adapting to diverse testing scenarios. To mitigate the impact of scale and orientation variations in objects on FSOD performance, Liu et al. [5] proposed a transformation-invariant network to learn the geometric invariance and spatial alignment features of support and query branches, which can help the detector cope well with the multiscale and arbitrary-orientation FSOD task. Moreover, Liu et al. [6] designed a label-consistent classifier and coarse-to-fine region proposal network (RPN) to alleviate the class, aspect ratio, and shape biases in diverse testing scenarios of FSOD.

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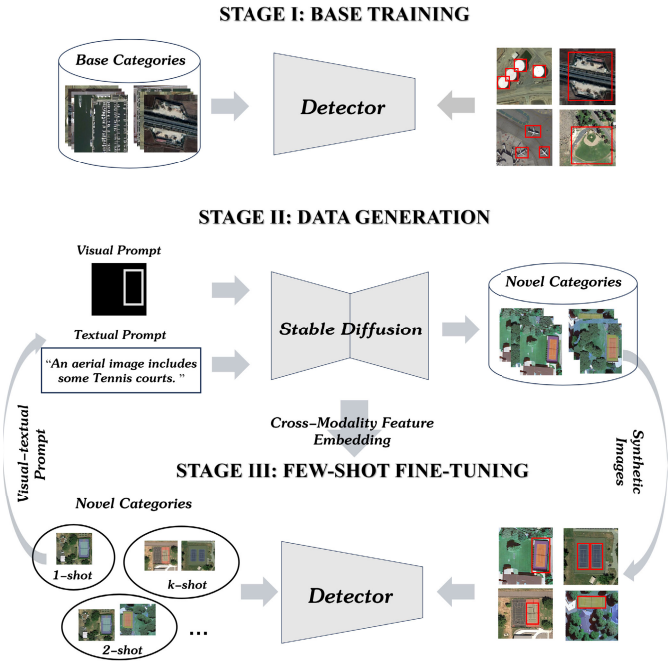


Fig. 1. Fine-tuning-based FSOD framework of the proposed CGK-FSOD.

Specifically, the agnostic bias related to class, scale, shape, or aspect ratio occurring in diverse testing scenarios should be carefully learned from extremely limited training samples, and they can be used to bridge the gaps of object characters between training and testing steps, while enabling the detector to generalize to unseen object detection. However, although previous studies [4], [5], [6] have demonstrated their effectiveness, only relying on extremely limited training samples to learn several agnostic biases is unilateral. Therefore, current works [4], [5], [6] can only obtain limited or unstable performance improvement and may have an enormous risk of failure in unseen object detection because of the presence of overfitting phenomenon and incomplete potential bias learning. In view of this, how to effectively avoid the overfitting phenomenon and learn more generalized concepts to deal well with potential agnostic biases (i.e., classes, scales, shapes, aspect ratios, and contextual information) becomes a very important problem to further improve FSOD ability. Recently, large-scale pretraining technology has made tremendous progress in the computer vision field, and it can develop universal domain-level knowledge to avoid overfitting in few-shot learning scenarios. Then, a series of powerful foundational models (e.g., contrastive language-image pretraining (CLIP) [8], SAM [9] and grounding DINO [10]) are pretrained and used for promoting the few- or zero-shot learning ability under various downstream tasks. For instance, CLIP [8] is pretrained on 400 million image–text pairs; thus, it can exhibit more impressive few- or zero-shot classification ability. Moreover, SAM [9] is pretrained on 1.1 billion pixel-level annotated masking data, and the model is also capable of achieving an excellent zero-shot segmentation ability. In addition, grounding DINO [10] is pretrained on a series of large-scale datasets, including object detection, grounding, and caption data. Through the designed large-scale grounded pretraining processes, a powerful few- or zero-shot learning ability would be endowed to grounding DINO [10]. In summary, the above-mentioned phenomena are very similar to those of the human cognitive system, which can achieve excellent few- or zero-shot learning ability owing to long-term knowledge learning and accumulation. Similarly, lengthy large-scale pretraining can assist foundational models in learning more universal domain-level knowledge, such as the process of knowledge learning and accumulation, which has been proven to be beneficial for few- or zero-shot learning. As a result, foundational models should be considered to be introduced into the few-shot learning framework, because they can encourage the model to further form more generalized concepts based on extremely limited training samples.

Therefore, to introduce universal domain-level knowledge for improving few-shot learning ability, while facilitating the establishment of more generalized concepts with the superior location-aware and identification abilities, a novel controllable generative knowledge-driven FSOD, named CGK-FSOD, is proposed for achieving unseen object detection from optical remote sensing imagery. Concretely, to expand the learnable data space of scarce training samples for preventing incomplete agnostic-bias learning, while avoiding the overfitting phenomenon, a visual–textual prompt-based data generation (VTP-DG) is designed to achieve high-quality sample generation that relies on pretrained foundational models (i.e., stable diffusion (SD) [11] and CLIP [8]). As shown in Fig. 1, the designed VTP-DG is considered to be plugged into stage II of the fine-tuning-based training procedure. It can not only bring more universal domain-level knowledge into the remote sensing domain but also provide controllable generated data, thereby avoiding the overfitting phenomenon and supporting more complete agnostic-bias learning according to stage III of few-shot fine-tuning. Furthermore, apart from more additional high-quality data generation, a series of cross-modality generative features in the latent representation space of the denoising generative process of SD are considered for reusing in feature representation learning at stage III of few-shot fine-tuning. Specifically, a cross-modality feature embedding (CMFE) is designed to integrate visual–textual prompt-guided generative features into the feature fusion step of the detector, which can enable feature representation learning scalability to adapt the detector well to agnostic bias in diverse testing scenarios. In addition, the CMFE can be incorporated with controllable data generation at stage III of few-shot fine-tuning to form more generalized concepts for FSOD. Finally, to demonstrate the effectiveness of the proposed CGK-FSOD, a large amount of comparison and ablation experiments are conducted on two public remote sensing benchmarks (e.g., DIOR [12] and NWPUVHR-10 [13]), and the results indicate that the proposed CGK-FSOD can achieve excellent FSOD performance.

1) CGK-FSOD is proposed for unseen object detection from optical remote sensing imagery. As well as we know, it is the first study to introduce pretrained foundational SD models from natural scene into remote sensing scene, which can be seamlessly integrated into a few-shot learning framework to form controllable data

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generation and scalable feature representation learning abilities.

2) To enrich the learnable data space of scarce training samples and prevent the incomplete agnostic-bias learning from very limited samples while avoiding overfitting phenomenon, a VTP-DG is designed for achieving diverse image synthesis based on pretrained foundational models (i.e., SD and CLIP). It can provide controllable object distributions for adapting the agnostic bias of unseen testing scenarios at the few-shot fine-tuning stage.
3) Except for extra high-quality training data generation, a CMFE is designed to reuse a series of cross-modality generative features in the latent representation space of the denoising generative process to enable the scalability of feature representation learning, which can integrate the visual–textual prompt-guided generative features into the feature fusion step of the detector and make the detector better adapt to agnostic bias from diverse testing scenarios.

II. Related Work

A. General Object Detection

Object detection has always been a critical research topic in the computer vision field, which requires locating and classifying the objects of random distribution within an image. Since the development of deep learning, a lot of deep-learning-based detectors have been proposed. According to their framework, they can be roughly divided into two types: two-stage detectors and one-stage detectors. Classic two-stage detectors usually contain Faster-RCNN [14], Mask-RCNN [15], and HTD [16]. These methods first predict the potential region proposals via an RPN and then further identify the category and regress parameters of bounding box. Unlike two-stage detectors, one-stage detectors such as YOLO series [17], [18], [19], RFB-Net [20] and CenterNet [21] directly predict the bounding boxes and categories on the basis of a set of anchor boxes or keypoints. These detectors have promoted the development of natural scene object detection. However, owing to the overlooking photographic perspective view of remote sensing imagery, these methods have shortcomings in detecting remote sensing objects with various scales, orientations, shapes, and aspect ratios against complicated background interference. For this purpose, many scholars are dedicated to improving the detection performance in various ways, such as the attention mechanism for feature optimization [22], [23], [24], anchor or keypoint sample refinement learning [25], [26], and multiscale feature fusion [27], [28], [29], [30]. Although great progress has been made, the above-mentioned methods still require a great deal of well-labeled images to support the model training step because they perform poorly under very low-data resource regimes, which is a critical problem for application in practical Earth observation systems. Therefore, in this work, our goal was to establish FSOD ability on the basis of a general object detector.

B. Few-Shot Object Detection

Different from the general object detection, FSOD is usually required in Earth observation to achieve unseen or novel object detection under the situation of scarce training samples. Then, to establish a better FSOD ability, many research efforts have been made, which can be roughly divided into two categories: fine-tuning-based [31], [32], [33], [34], [35], [36] and meta-learning-based [37], [38], [39], [40], [41], [42], [43], [44] strategies. First, with respect to the fine-tuning-based strategy, the object detector is pretrained on a large-scale dataset, and then only a few training samples are required for the model fine-tuning step to achieve unseen object detection. For example, Chen et al. [31] introduced regularized transfer learning for FSOD, which can facilitate the learning of object detection knowledge from the source and target domains; thus, it can achieve unseen object detection by fine-tuning on very limited training samples from the target domain. Han and Lim [36] proposed a visual–language few-shot object detector according to modern foundation models (i.e., the contrastive pretrained DINOv2 [45] and large language model [46]), because modern foundation models can provide more generalized domain-level knowledge to facilitate unseen object detection by fine-tuning a model only relying on a few training samples. In addition to fine-tuning-based methods, there are several studies focusing on meta-learning-based methods for FSOD. These methods use episodic training tasks to establish FSOD ability for coping with diverse few-shot testing scenarios. Specifically, Wang et al. [37] proposed a meta-learning-based approach to learn class-agnostic and class-specific parameters in a unified way. Meta-RCNN [39] was set up based on episodic learning, which would make the detector familiarize itself with FSOD tasks while setting up more generalized concepts for unseen object detection. Guo et al. [40] designed a meta-learning-based FSOD approach to learn scale-aware and discriminative prototypes from very limited training samples. In addition, many recent works [47], [48] are focusing on more challenging cross-domain few-shot object detection (FSOD), aiming to tackle the domain gap. However, for both the fine-tuning-based [31], [33], [34], [35], [36] and meta-learning-based methods [37], [39], [40], [41], [42], [43], [44], when very limited training samples are used for model fine-tuning or meta-training, there is a very enormous risk of overfitting or incomplete agnostic-bias learning, which would easily present inferior or unstable FSOD performance. Thereby, in this study, how to enrich the learnable data space and be incorporated with scalable feature representation learning for forming more generalized FSOD concepts needs to be further studied.

C. Data Generation Strategy

Data generation strategy is a very hot research topic, which has been widely studied in the computer vision field. This task is dedicated to improving the quality of data generation, such as image reconstruction [49], [50], [51], style transfer [52], [53], [54], [55], image editing [56], [57], [58], and data augmentation [59], [60], [61]. In general, the generative methods can be roughly summarized into two categories:

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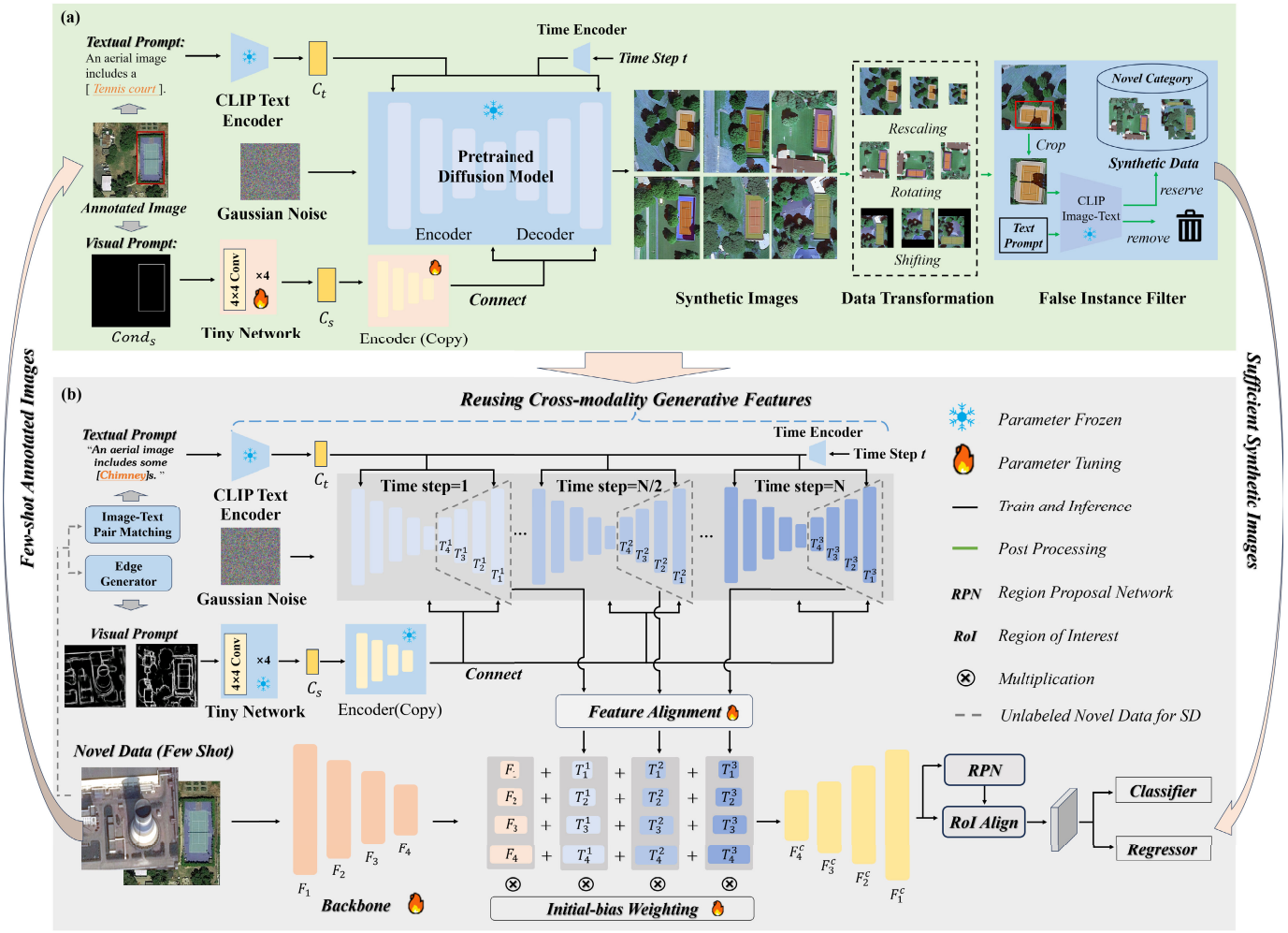


Fig. 2. Architecture and framework of the proposed CGK-FSOD. It includes two parts (a) visual–textual prompt-based data generation (VTP-DG) and (b) CMFE for FSOD.

pixel-level and latent-space-based generation models. First, the representative pixel-level generative model is the generative adversarial networks (GANs), which requires to learn the pixel-level projection from an initial distribution to any distribution based on an adversarial generation mechanism. For example, CycleGAN [62] was built to learn mapping from an input image to an output image via the aligned image pairs and cycle consistency. Subsequently, conditional adversarial networks are designed by the same generic approach to address the problem of image-to-image translation, which can further improve the effect of pixel-level generation. However, the images generated in pixel-level space usually have drawbacks of slow inference and high training costs. Although these drawbacks have been partially addressed by sampling strategies [63], [64] and hierarchical approaches [65], training the model on very high-resolution images still requires expensive gradient calculations. Recently, diffusion denoising probabilistic models [66] have emerged, which have become a new trend in the field of image generation because of their remarkable generative ability according to latent space. Where, by conditioning text prompts into latent representation space for training the denoising autoencoder, it can avoid the drawbacks of pixel-level-based generation models. Hence, the text-to-image diffusion models have attracted considerable attention. Second, regarding latent-space-based generation way, training diffusion models on latent space [67], [68] allows for the consumption of limited computational resources while retaining their quality and flexibility. Specifically, DALL-E2 [69] adopted the CLIP text encoder but inverted its image encoder with a diffusion model for generating realistic images from the latent space of CLIP. In addition, a milestone method that trains diffusion on latent space is SD [67], which reaches out to a large-scale text-to-image diffusion model by training on LAION-5B [70]. Despite great progress, text-to-image generation methods usually fail to depict controllable spatial distributions of images. To tackle this problem, recent progresses [11], [71], [72], [73], [74], [75], [76], [77] have been made in controllable text-to-image generation. For examples, Avrahami et al. [73] used semantic segmentation results to drive the text-to-image generation with sparse scene control. Then, Zhang et al. [11] designed spatially localized input conditions (i.e., Canny edges, human keypoints, cartoon line drawings) to pretrained text-to-image diffusion for controllable generation. To further improve fine-tuning efficiency and preserve the subject fidelity, Li et al. [77] designed a subject-driven image generation model that

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supports multimodal control to consume inputs of subject images and text prompts, whose multimodal encoder is pretrained to provide subject representation. Related to the above-mentioned studies, our research was aiming to construct an object detection sample generation method and introduce it into the FSOD framework to enrich the learnable data and set up scalable feature representation for FSOD promotion.

III. Methodology

A. Overview

1) Problem Definition: Following the general FSOD definition, the dataset for FSOD task should be divided into two parts: the base categories C_{base} and novel categories C_{novel}. Specifically, the base categories are composed of a relatively large-scale dataset $D_{\text{base}} = \{(x_i, y_i), y_i \in C_{\text{base}}\}_{i=1}^{N}$ with abundant annotated objects. In contrast, the novel categories consist of a small-scale dataset $D_{\text{novel}} = \{(x_i, y_i), y_i \in C_{\text{novel}}\}_{i=1}^{K}$ with only K (i.e., $K = 1, 2, 3, \ldots, 20$) annotated object samples per category, where x represents the images and y denotes their corresponding annotations. In addition, the categories in C_{base} and C_{novel} are nonoverlapping, namely, $C_{\text{base}} \cap C_{\text{novel}} = \emptyset$. In general, as shown in Fig. 1, D_{base} would be adopted for base detector training at stage I, which can make the detector acquire a powerful generalization ability to detect foreground objects. Before stage III of few-shot fine-tuning, each category of C_{novel} is sampled with K shot samples to prepare for data generation at stage II, as illustrated in Fig. 1. After that, the generated data and features can be introduced into the FSOD framework to set up controllable data generation and scalable feature representation learning abilities. Subsequently, several synthetic data $D^*_{\text{novel}} = \{(x_i^*, y_i^*), y_i^* \in C_{\text{novel}}\}_{i=1}^{L}$ would be generated from the already sampled K shot samples of each novel category, where L represents the number of synthetic images. Finally, the union set $D^*_{\text{novel}} \cup D_{\text{novel}}$ is adopted for stage III of few-shot fine-tuning. In general, it is a challenge for FSOD to make the detector avoid overfitting and prevent incomplete agnostic-bias learning from very limited training samples. Thus, it is worth exploring how to establish a good controllable data generation capability at stage II and build scalable feature representation learning that incorporates controllable data generation at stage III of few-shot fine-tuning to set up more generalized concepts for FSOD.

2) Overall Pipeline: As shown in Fig. 1, the CGK-FSOD is proposed for establishing the abilities of controllable data generation and scalable feature representation learning, which can assist the object detector in learning more generalized concepts from very limited training samples to cope well with agnostic bias from diverse testing scenarios of FSOD. The details of the proposed CGK-FSOD are illustrated in Fig. 2, which consist of two aspects, i.e., the designed VTP-DG and CMFE. First, regarding the designed VTP-DG, the pretrained foundational models from natural scenes (i.e., the CLIP and text-to-image SD) would be fine-tuned into the remote sensing domain via a carefully designed visual–textual prompt. Specifically, as illustrated in Fig. 2(a), the labeled name for each category can be seen as a textual prompt, and then it can be fed into the frozen text encoder of CLIP for producing a textual vector. Moreover, the spatial layout distribution of labeled bounding boxes (bboxes) that correspond to textual prompts would be considered as visual prompts. Next, they are further fed into a learnable tiny network and copied encoder of SD, and then the multiscale visual features can be obtained. Subsequently, these prepared textual vectors and multiscale visual features are simultaneously conditioned on the pretrained SD model to control the denoising generative process depending on the two inputs of Gaussian noise and different time steps t. After that, the generalized domain-level knowledge in the natural scenes would be transferred into the remote sensing domain to establish a good controllable image generation ability for novel categories. As illustrated in Fig. 2(a), the already sampled K-shot training samples per class can be used to generate their corresponding visual–textual prompt for additional synthetic image generation. Besides, the following steps of data transformation and false instance filtering are also designed to further enrich the learnable data space while ensuring the generative quality of synthetic images. Following this way, the overfitting phenomenon can be effectively alleviated, and the incomplete agnostic-bias learning from very limited training samples can be prevented. Second, except for controllable high-quality data generation, the feature representation learning of FSOD can be improved by the designed CMFE at stage III of few-shot fine-tuning. As shown in Fig. 2(b), corresponding to different time steps of latent representation space, there are a series of cross-domain generative features within the denoising generative process of SD that can be reused in the feature fusion step of the detector. Notably, the initial-bias weighting operation is designed to achieve the CMFE because it not only can integrate features with diverse generative abilities into the detector, which can increase the scalability of feature representation under few-shot learning, but also make it be incorporated with controllable data generation to form more generalized concepts, and then adapting well to agnostic bias from diverse testing scenarios of FSOD. Next, the designed VTP-DG and CMFE are elaborated as follows.

B. Visual–Textual Prompt-Based Data Generation

As shown in Fig. 2(a), the parameters of the pretrained CLIP and SD models are first frozen. Then, according to the encoder–decoder architecture of the SD model, several extra learnable parameters of the tiny network and trainable copy of encoder are used for SD model fine-tuning, which can transfer the generalized domain-level knowledge from natural scenes into the remote sensing scene for data generation task. Therefore, a large-scale public remote sensing object detection dataset $D_{\text{rs}} = \{(x_i^{\text{rs}}, y_i^{\text{rs}}), y_i^{\text{rs}} \in C_{\text{rs}}\}$ called DOTA [78] is used for fine-tuning the pretrained SD model, where C_{rs} indicates the categories of DOTA. These known categories and their corresponding bboxes from D_{rs} are regarded as visual–textual prompts. Notably, for the textual prompt, the provided category labels C_{rs} are inserted into the predefined textual templates Temp_t, i.e., "*An aerial image includes* [num_1] [class_1] *and* [num_2] [class_2], ..., [num_n] [class_n]." Here, num_n denotes

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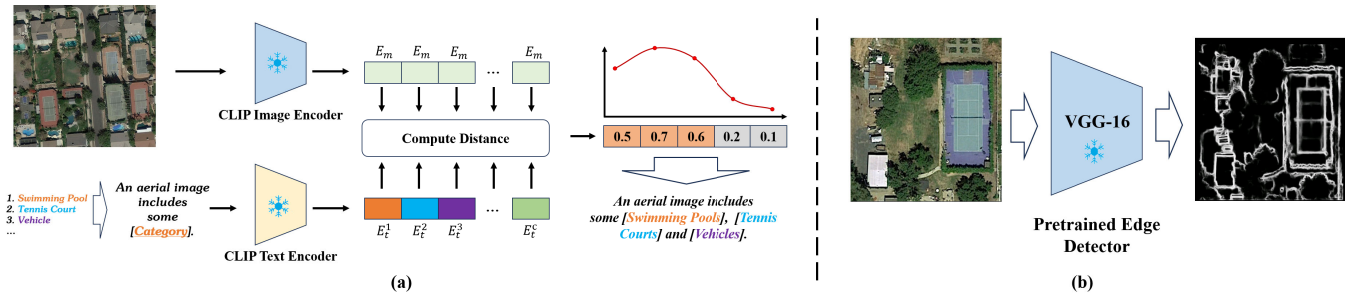


Fig. 3. Acquisition details of visual and textual prompts in the proposed CMFE. (a) Process of obtaining the textual prompts by image–text pair matching. (b) Process of generating visual prompts via a pretrained edge detector.

the number of objects, n is the index, and class_n represents the nth category. Next, the frozen textual encoder of the pretrained CLIP is used to encode Temp_t as textual vector C_t, which can be expressed as

$$C_t = \text{FrozenClipTextEncoder}\left(\text{Temp}_t\right). \tag{1}$$

After (1), C_t is prepared for conditioning on the frozen encoder–decoder of the pretrained SD model. Besides, related to the visual prompt, which corresponds to the textual prompt, the bboxes $y_i^{\text{rs}} = (a_i, b_i, w_i, h_i)$ are drawn as white rectangles on the black background mask and it can be seen as the visual prompt called Cond_s, where (a_i, b_i) denotes the center point coordinates of bboxes and (w_i, h_i) represents the width and height information of bboxes. Subsequently, a visual prompt Cond_s is fed into a learnable tiny network, which consists of four convolution layers with 4×4 kernels and 2×2 strides to project the visual prompt as visual vector C_s, which can be expressed as

$$C_s = \text{TinyNetwork}(\text{ConvLayer}(\text{Cond}_s)). \tag{2}$$

After (2), the obtained visual vector C_s would be fed into a learnable encoder, which is copied from the frozen encoder of the pretrained SD model, to learn the multiscale spatial distributions. Then, the textual vector C_t and output features of the learnable encoder are conditioned on the frozen encoder–decoder of the pretrained SD model. Specifically, a cross-attention mechanism is adopted for conditioning the textual vector C_t into the entire encoder–decoder. Moreover, the output features of the learnable encoder are concatenated with the output features from the decoder. These processes allow for the denoising data generation process to be controlled on the basis of the inputs of Gaussian noise and different time steps by the following equation:

$$\text{DG}_{\text{SynImg}} = \text{SD}(x|C_t, \text{encoder}(C_s), z_t, t, \varepsilon \sim \mathcal{N}(0, 1)). \tag{3}$$

In (3), $\varepsilon \sim \mathcal{N}(0, 1)$ is Gaussian noise, and z_t indicates several latent cross-modality features of $\text{SD}(\cdot)$ for denoising data generation on the basis of different input time steps t. Next, according to the conditioning of C_t and $\text{encoder}(C_s)$, a series of synthetic images $\text{DG}_{\text{SynImg}}$ can be generated. Accordingly, after generalized domain-level knowledge is transferred from the natural scene to the remote sensing domain for data generation according to D_{rs}, the SD model incorporating CLIP can be used to achieve controllable data generation for remote sensing object detection. Therefore, as shown by the left arrow in Fig. 2, a few-shot annotated training image of novel categories that is used for few-shot fine-tuning of stage III can be expanded by the designed VTP-DG. It can provide extra synthetic images with various object appearances and background distributions. This is beneficial for enriching the learnable data space in few-shot fine-tuning. Furthermore, following by the synthetic image generation, the data transformation (i.e., rescaling, rotating, and shifting) and false instance filter are also designed to further enlarge the learnable data space while ensuring training data quality for few-shot fine-tuning. The scale range of rescaling is 0.6–1.4, the rotation of angles are 0°, 90°, 180°, and 270°, and the image shifts are the default as randomly shifting synthetic images within 1/4 of the image size referring to the center point of images. After that, there should be rich scale, orientation, and position shifting information in the synthetic images, and they can be used for setting up a sufficient data representation learning space to avoid the overfitting phenomenon and prevent incomplete agnostic-bias learning from very limited training samples. Inevitably, among generative and augmented synthetic images, some unreasonable or low-quality region-level annotated object instances exist. Consequently, to avoid low-quality synthetic image samples misleading few-shot fine-tuning with novel categories, several object instances existing in synthetic image samples are cropped and encoded by the image encoder of CLIP. Subsequently, the textual template, i.e., "*This is an image of [class]*" that is filled in with the corresponding category label is also fed into the textual encoder of CLIP. Next, the score of the image-to-textual pair from CLIP is used to determine to whether remove or retain the synthetic image samples for few-shot fine-tuning.

C. Cross-Modality Feature Embedding

As mentioned in Section III-B, the generative SD model can implicitly learn semantic relationships between low-level visual and high-level textual concepts from large-scale image–text pairs through the use of a denoising autoencoder, thereby generating diverse high-quality controllable data. Apparently, the excellent image generation ability benefits from the powerful latent cross-modality generative features within the denoising generative process. Inspired by this, we try to consider a series of latent cross-modality generative

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TABLE I
BASE–NOVEL CATEGORIES SPLIT SETTING

Dataset	Split	Novel Categories					Base Categories
DIOR	1	baseball field	basketball court	bridge	chimney	ship	rest
	2	airplane	airport	expressway toll station	harbor	ground track field	rest
	3	dam	golf course	storage tank	tennis court	vehicle	rest
	4	expressway service area	overpass	stadium	train station	windmill	rest
NV10	/	airplane		vehicle		tennis court	rest

features to make up for the partial semantic features extracted from the conventional backbone of the FSOD detector on extremely limited training samples.

In theory, the generative denoising process is endowed with an imaginative ability and helps capture interesting structure from input distribution [79], [80], which allows for providing creative and robust feature representation for few-shot learning. In addition, the visual–textual prompt-guided generative features within denoising process can offer a spatial layout prior and facilitate a general understanding of image content, which can promote the location-aware and identification abilities for FSOD task. To fully use these benefits to increase the scalability of feature representation learning, a new CMFE is proposed to integrate the latent cross-modality generative features into the detector and make the detector effectively adapt to agnostic bias in diverse testing scenarios. Formally, this process is modeled as the distribution $P(y|x, C)$, where y is the predicted result; x is the input image; and C denotes a set containing all the category names. Then, $P(y|x, C)$ is formulated as below

$$P(y|x, C) = P(y|F)P(F|x, C_t, C_s)P(C_t|x, C)P(C_s|x) \quad (4)$$

where F denotes a set of latent cross-modality features; and C_t and C_s denote the textual vector and visual vector, respectively. Next, we would further illustrate each term of (4) and its details.

As shown in the right arrow in Fig. 2, after stage II of data generation, sufficient synthetic images combined with the existing limited training images are applied for the few-shot fine-tuning of stage III. Under the process of few-shot fine-tuning, first, $P(C_t|x, C)$ is used to extract the textual vector C_t. Considering that the category of objects in the testing scenarios is unknown, the pretrained CLIP [8] is used for encoding the input image x as image feature E_m to match the text prompts. And then, the text encoder of CLIP [8] is used for encoding the inserted textual template with a set of category names, i.e., "*An aerial image includes some* [class_1, class_2, ..., class_c]." as text feature $E_t^c, c \in C$, which contains several potential object category names. The image–text pair matching can be measured by the cosine similarity D_c between E_m and E_t^c as follows:

$$D_c = \left(E_m E_t^c\right) / \left(\|E_m\| \left\|E_t^c\right\|\right). \quad (5)$$

After (5), when the cosine similarity D_c is greater than a determined threshold, the corresponding class names would be chosen to consist of textual prompts, which represent the content of the input images, as shown in Fig. 3(a). And then, the retrieved text prompts are fed into the text encoder of CLIP to get the textual vector C_t. Second, $P(C_s|x)$ is responsible to extract the visual vector C_s. Here, the pretrained model [81] is adopted for producing binary edge detection results, where the occlusive binary edges can be viewed as visual prompts that may contain potential objects, as shown in Fig. 3(b). Next, the visual prompts are fed into the frozen tiny network to obtain the visual vector C_s. Subsequently, $P(F|x, C_t, C_s)$ represents that high-quality synthetic images x and autogenerated vectors C_s and C_t from visual–textual prompts are input into the backbone network of the detector and the frozen encoder–decoder of the SD model, respectively. As shown in Fig. 2(b), multiscale feature maps $F_i, i \in [1, 4]$ can be acquired from the backbone network of the detector, and then the latent cross-modality generative features $T_i^j, i \in [1, 4], j \in [1, 3]$ that correspond to different time steps (i.e., $t = 1, N/2, N$) of the denoising generative process can also be acquired for embedding into F_i, which can bring more diverse cross-modality generative feature embedding into basic multiscale feature extraction to enable the object detector to learn potential representation biases. To reasonably integrate them, an initial-bias weighting operation is designed for achieving CMFE. In details, the sampled latent cross-modality generative features $T_i^j, i \in [1, 4], j \in [1, 3]$ are fed into one convolution layer to align their feature dimensions with multiscale feature maps $F_i, i \in [1, 4]$. The convolution layer is composed of 3×3 and 1×1 convolutions. The designed initial-bias weighting operation is subsequently applied for $T_i^j, i \in [1, 4], j \in [1, 3]$ and $F_i, i \in [1, 4]$, which can adaptively select beneficial information of CMFE among multiscale feature maps as follows:

$$F_i^c = W_1 F_i + W_2 T_i^1 + W_3 T_i^2 + W_4 T_i^3. \quad (6)$$

In (6), $F_i^c, i \in [1, 4]$ represents the embedded feature representation, and then, the trainable weighting factor W_i, $i \in [1, 4]$ can be expressed as

$$W_i = e^{z_i} \Big/ \sum_{i=1}^{4} e^{z_i}, \quad i \in [1, 4]. \quad (7)$$

In (7), z_1, z_2, z_3, z_4 are initialized as $[10, 0, 0, 0]$, which aims to prevent the negative impact of cross-modality generative features on pretrained detectors at the initial stage of few-shot fine-tuning. Finally, in the form of generative knowledge, the embedded cross-modality features are used for subsequent classification and regression tasks for FSOD. In general, the use of high-quality synthetic images incorporated with CMFE can effectively form more generalized concepts for FSOD at

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stage III of few-shot fine-tuning. After finishing stages II and III, the few-shot object detector can achieve unseen object detection on the basis of fine-tuning architecture, as illustrated in Fig. 2(b).

D. Loss Function

With respect to stages II and III, which are presented in Fig. 1(a) and (b), two different loss functions are adopted for data generation and few-shot fine-tuning. Therefore, the loss functions of the designed VTP-DG and CMFE-based few-shot fine-tuning are elaborated below.

1) Data Generation: As described previously, the designed VTP-DG is constructed to transfer the generalized domain-level knowledge of large-scale pretrained CLIP and SD models into the remote sensing domain. The pretrained CLIP is used to assist in setting up visual–textual prompts and matching pairs in the remote sensing domain, and the task of data generation depends mainly on the pretrained SD model. Hence, the loss function for data generation can be expressed as follows:

$$\mathcal{L}_{\mathrm{DG}} = \mathbb{E}_{z_0,t,C_t,C_s,\varepsilon\sim N(0,1)}\left[\|\varepsilon - \varepsilon_\theta(z_t, t, C_t, C_s)\|_2^2\right]. \quad (8)$$

In (8), $\mathcal{L}_{\mathrm{DG}}$ represents the total loss of data generation; its goal is to learn a network parameter ε_θ to predict the noise ε, which is progressively added to an input image z_0. Meanwhile, the noisy image z_t can be produced as potential representations to complete the data generation task according to different time steps and various conditions C_t and C_s.

2) Few-Shot Fine-Tuning: At stage III of few-shot fine-tuning, some synthetic images with their corresponding labels from stage II are combined with limited few-shot annotated training images, and they can be used for fine-tuning the pretrained detector of stage I followed by the loss function of [34]. The total loss function is shown in the following equation:

$$\mathcal{L}_{\mathrm{obj}} = \mathcal{L}_{\mathrm{RPN}} + \mathcal{L}_{\mathrm{RoI}} \quad (9)$$

where $\mathcal{L}_{\mathrm{obj}}$ is the total loss for the fine-tuning detector. $\mathcal{L}_{\mathrm{RPN}}$ represents the loss function of the RPN, as follows:

$$L_{\mathrm{RPN}} = \frac{1}{N_{\mathrm{obj}} + M_{\mathrm{obj}}} \sum_{i=1}^{N_{\mathrm{obj}}+M_{\mathrm{obj}}} L^i_{B\mathrm{cls}} + \frac{1}{N_{\mathrm{obj}}} \sum_{i=1}^{N_{\mathrm{obj}}} L^i_{P_{\mathrm{reg}}}. \quad (10)$$

In (10), N_{obj} is the total number of chosen anchors. M_{obj} is the number of selected positive anchor samples for refinement. When the ith anchor is in the mini-batch, $L^i_{B\mathrm{cls}}$ represents the binary cross-entropy loss which is used for classifying the foreground objects and $L^i_{P\mathrm{reg}}$ denotes the smooth L_1 loss which can be used for calculating the localization loss of bboxes. In addition, $\mathcal{L}_{\mathrm{RoI}}$ in (9) represents the loss function of the region-of-interest (RoI) prediction head, which is shown below

$$L_{\mathrm{RoI}} = \frac{1}{N_{\mathrm{RoI}}} \sum_{i=1}^{N_{\mathrm{RoI}}} L^i_{K\mathrm{cls}} + \frac{\lambda}{M_{\mathrm{RoI}}} \sum_{i=1}^{M_{\mathrm{RoI}}} L^i_{K\mathrm{cls}} + \frac{1}{N_{\mathrm{RoI}}} \sum_{i=1}^{N_{\mathrm{RoI}}} L^i_{R\mathrm{reg}} \quad (11)$$

where N_{RoI} is the number of RoIs in a mini-batch. $L^i_{K\mathrm{cls}}$ is the cross-entropy loss over all categories. M_{RoI} represents the number of selected RoIs in [34]. $L^i_{P\mathrm{reg}}$ also denotes the smooth L_1 loss. Different from previous FSOD studies, owing to the controllable data generation capability at stage II, it is possible to generate sufficient high-quality labeled samples to support few-shot fine-tuning.

IV. Experiments and Analysis

A. Datasets and Implementation Details

1) Dataset Description: To demonstrate the effectiveness of the proposed CGK-FSOD, two public remote sensing datasets, DIOR [12] and NWPUVHR-10 (NV10) [13], are selected for assessing FSOD performance.

DIOR [12] is a large public remote sensing object detection dataset that contains 23 463 images with 192 472 instances involving 20 common categories. The spatial resolution ranges from 0.5 to 30 m, and the size of each image is 800 × 800. Its categories include windmill, tennis court, baseball field, vehicle, stadium, ground track field, airport, overpass, storage tank, harbor, ship, bridge, basketball court, expressway service area, golf course, airplane, dam, train station, expressway toll station, and chimney. To split the dataset, 11 724 images are selected as the trainval set, and the remaining images are used for the test set. In addition, to establish the setting of FSOD, five categories are chosen as novel categories, and the remaining 15 categories are regarded as base categories. Specifically, following the setting of [82], four different category splits are constructed to verify robustness, as depicted in Table I.

NV10 [13] is a multiclass remote sensing object detection dataset that contains 650 available images with annotations. Among them, 455 images are used as the trainval set, and the remaining 195 images are used for the test set. Due to its irregular image size, the original images are cropped to 800 × 800 with 200 overlaps. Moreover, the dataset contains 10 categories, including the airplane, ship, storage tank, baseball diamond, tennis court, basketball court, ground track field, harbor, bridge, and vehicle. To simulate the few-shot setting, as shown in Table I, the airplane, vehicle, and tennis court are selected as novel categories, and the rest of the seven categories are referred to as base categories.

2) Implementation Details: Our proposed method is based on the plain FSOD detector, i.e., MPSR [34]. Actually, owing to flexible designing of our proposed CGK-FSOD, they can be compatible with other FSOD detectors. And then, ResNet-101 [88] is selected as the backbone network that is pretrained on ImageNet, and then the pretrained SD model adopts the SD [67] which is pretrained on LAION-5B [70]. To better provide the image–text pair matching for remote sensing image, the pretrained CLIP model uses RemoteCLIP [89] which is trained on a large-scale remote sensing dataset. Specifically, the architecture of the selected CLIP model is ViT-Base-32 of 87 M parameters. In addition, following the paradigm of fine-tuning-based strategy, the proposed CGK-FSOD contains three phases, as shown in Fig. 1. Regarding base training of stage I, we follow the default setting of

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TABLE II

COMPARISON OF FSOD PERFORMANCE ON DIOR [12] UNDER THE SETTINGS OF 3, 5, 10, AND 20 SHOTS. THE BEST AND SECOND BEST RESULTS (mAP) ARE MARKED WITH BOLD AND UNDERLINE, RESPECTIVELY

Split	Setting	RepMet [43]	TFA [33]	Meta-RCNN [39]	MPSR [34]	P-CNN [82]	SAE-FSDet [6]	FPD [83]	SAGS [84]	MSOCL [85]	CGK-FSOD
1	3-shot	11.3	11.6	14.0	17.5	18.0	26.9	12.4	29.3	28.5	**41.6**
	5-shot	11.3	16.2	16.8	26.9	22.8	32.6	16.7	31.6	34.6	**42.0**
	10-shot	12.1	22.9	22.0	28.0	27.6	37.3	18.5	36.4	40.8	**44.7**
	20-shot	12.2	23.7	24.3	30.1	29.6	42.2	16.6	40.2	43.3	**47.5**
2	3-shot	5.2	0.0	10.5	7.4	14.5	8.1	4.8	12.6	16.8	**26.0**
	5-shot	5.6	3.6	12.5	21.3	14.9	9.5	5.8	15.5	20.6	**26.0**
	10-shot	5.9	5.2	15.3	21.8	18.9	18.7	5.3	19.7	23.6	**24.8**
	20-shot	6.8	7.8	20.5	27.0	22.8	26.2	7.9	23.8	**29.2**	28.7
3	3-shot	5.5	11.1	12.2	7.8	16.5	20.6	5.3	**20.9**	19.6	18.0
	5-shot	5.7	13.8	15.4	18.6	18.8	20.6	9.5	**24.8**	21.4	22.6
	10-shot	7.4	14.6	20.3	19.2	23.3	27.3	10.2	**28.6**	26.0	26.7
	20-shot	7.8	18.9	26.3	21.4	28.8	32.7	15.2	**36.1**	30.2	33.1
4	3-shot	10.9	9.4	11.6	7.8	15.2	8.4	6.4	17.5	15.6	**17.9**
	5-shot	11.6	9.6	13.6	21.2	17.5	13.4	7.7	19.7	19.0	**23.8**
	10-shot	12.8	11.4	17.0	24.0	18.9	**28.3**	10.4	23.7	24.6	24.8
	20-shot	16.7	12.8	22.2	30.9	25.7	28.6	11.2	27.7	29.4	**31.3**

TABLE III

COMPARISON OF FSOD PERFORMANCE ON NV10 [13] UNDER THE SETTINGS OF 3, 5, 10, AND 20 SHOTS. THE BEST AND SECOND BEST RESULTS (mAP) ARE MARKED WITH BOLD AND UNDERLINE, RESPECTIVELY

Setting	Category	TFA [33]	Meta-RCNN [39]	MPSR [34]	SAE-FSDet [6]	FsDetView [86]	FPD [83]	FSCE [87]	CGK-FSOD
3-shot	Airplane	0.0	3.1	12.4	2.7	0.5	3.0	4.0	**18.9**
	Vehicle	4.9	5.3	11.2	0.2	6.6	9.3	7.3	**42.1**
	Tennis court	5.3	1.7	36.6	12.7	2.9	1.1	35.2	**60.1**
	mAP	3.4	3.4	20.1	5.2	3.3	4.4	15.5	**40.4**
5-shot	Airplane	0.0	0.9	**32.2**	13.2	0.3	10.1	8.8	22.0
	Vehicle	6.1	8.9	13.6	0.3	7.3	9.0	5.8	**40.5**
	Tennis court	6.8	2.3	29.9	25.4	4.7	12.3	30.0	**62.2**
	mAP	4.3	4.1	25.2	13.0	4.1	10.4	14.9	**41.6**
10-shot	Airplane	0.0	0.8	37.1	**40.6**	1.2	24.0	21.2	21.0
	Vehicle	4.9	11.9	32.1	17.6	7.2	20.7	26.9	**45.6**
	Tennis court	7.5	2.7	31.8	33.0	4.5	25.7	40.8	**66.8**
	mAP	4.1	5.2	33.7	30.4	4.3	23.5	29.6	**44.5**
20-shot	Airplane	0.0	0.0	46.3	**47.3**	32.2	0.1	41.7	38.2
	Vehicle	5.4	17.1	**54.5**	33.5	7.5	41.7	43.3	44.5
	Tennis court	9.1	1.8	7.5	39.6	4.8	9.1	9.7	**62.1**
	mAP	4.8	6.3	36.1	40.1	4.2	27.7	31.6	**48.3**

MPSR [34] to train the detector according to base categories. Thus, the detector is trained by 80 000 iterations, and the initial learning rate is set as 0.02 and it decreases by half at 56 000 and 70 000 iterations. At stage III of few-shot fine-tuning, we replace the detector head of the trained base detector and fine-tune it relying on few-shot data for 4000 iterations. The learning rate is also set as 0.02, which decreases by half at 2800 and 3500 iterations. Then, the batch size of two phases is specified as 2. To simulate the few-shot learning scenarios, we separately select $3, 5, 10$, and 20 annotated instances for each category from the training set. Specifically, at stage II of data generation, because our proposed CGK-FSOD can finely generate the synthetic samples according to visual–textual prompt, the combination of synthetic data and corresponding few-shot data is used for jointly fine-tuning the base detector. Subsequently, as described in the Section III-B, the image–text pair matching score from CLIP is designed to filter the false annotated instances, where the threshold is empirically set as 0.2. Next, the data generation process solely depends on the existing few-shot annotation information or self-made bounding box and category, without extra information leakage. The number L of synthetic images is set as 100 for $3, 5, 10$, and 20-shots. Then, the hyperparameters about the stage II of data generation are illustrated below. First, SD V2.1 [67] is introduced as the foundational model for valid data generation. To transfer universal domain-level knowledge from foundational model into the remote sensing domain, we further fine-tune the model by additional designed visual–textual prompts from remote sensing imagery. About the implementation details, the batch size is set as 4 and the learning rate is initialized as 1e-5. The time step N of the denosing process is 1000. As mentioned above, the pretrained diffusion model and text encoder of CLIP are frozen, and then only the learnable tiny network and trainable copy of encoder

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for visual prompt are fine-tuned on remote sensing data. At the inference phase, the sampling steps are designated to 50. For more implementation details, refer to [11].

Moreover, the experimental environment is built on PyTorch framework and two NVIDIA GeForce RTX A6000 GPUs. For evaluation metrics, we follow the PASCAL VOC 2007 benchmark [90] and adopt the average precision (AP) and mean AP (mAP) to evaluate the performance. The intersection over union (IoU) between the predicted results and ground truths is set to 0.5 by default.

B. Comparison Analysis

In this section, all-round comparison experiments are conducted on the DIOR [12] and NV10 [13] datasets. The results are reported in Tables II and III. Next, to fully demonstrate the superiority of our proposed CGK-FSOD, a detailed analysis is provided below.

1) Results on DIOR Dataset: Under the challenging large-scale remote sensing object detection dataset DIOR [12], nine state-of-the-art (SOTA) FSOD methods are selected for comparison, including meta-learning-based methods (i.e., RepMet [43], Meta-RCNN [39], P-CNN [82], FPD [83], and SAGS [84]) and fine-tuning-based methods (i.e., TFA [33], SAE-FSDet [6], MSOCL [85], and MPSR [34]). Some comparisons [33], [34], [39], [43] are designed from natural scenes, and other methods [6], [82], [83], [84], [85] are all commonly designed for the remote sensing imagery. The comparison results are reported in Table II. Specifically, in Table II, the first column depicts the setting of category split, and it is corresponding to those in the second column of Table I. The second column in Table II denotes the number of annotated samples (i.e., K-shot) in the novel categories for training. And then, the 3rd–12th columns denote the reported results (i.e., mAP of novel categories) from different FSOD detectors.

According to the last column of Table II, the proposed CGK-FSOD achieves four SOTA results under the setting of split 1, three SOTA results and one suboptimal result under the setting of split 2, two suboptimal results under the setting of split 3 and three SOTA results, and one suboptimal result under the setting of split 4. These results indicate that the proposed CGK-FSOD can effectively address diverse testing scenarios and achieve superior FSOD performance. Next, a more detailed comparison analysis would be provided. As shown in the 3rd–5th columns of Table II, the methods from the natural scene achieve poor FSOD performance compared with most methods from the remote sensing domain. The reason for this is their inability to adequately adapt to the more intricate scale variations in remote sensing objects, which stems from the scale-bias inherent in the limited number of annotated samples. This issue can be explained by the following results. Under the setting of split 2 (i.e., the 6th–9th lines of Table II), the results are evaluated in five categories including the airport, ground track field, airplane, expressway toll station, and harbor. Obviously, a significant scale difference clearly exists among these categories. In this situation, the nature scene methods of RepMet [43] and TFA [33] perform extremely poorly, even with a result of zero in the 4th column of Table II. The results indicate the shortcomings of the detectors in detecting unseen objects with a large-scale variance. However, MPSR [34] shows a significant performance gain compared with other detectors that are designed for natural scene by proposing a multiscale positive sample refinement to enlarge the learning space of object scale distributions. In addition, as shown in the 11th column of Table II, the remote sensing scene method of MSOCL [85] exhibits superior performance under the setting of split 2, which can achieve two second best results (i.e., 16.8% mAP in three-shot and 23.6% mAP in ten-shot) and one best result (i.e., 29.2% mAP in 20-shot) because the designed multiscale object contrastive learning can effectively alleviate the impact of scale bias for FSOD. The above analysis illustrates that considering the scale bias between few-shot training data and diverse testing data is very important, and expanding object scale distributions for few-shot learning is an essential strategy for improving FSOD ability in the remote sensing domain. Despite this progress, the above FSOD methods, whether designed for natural or optical remote sensing images, still stick in depending on extremely limited annotated samples. In fact, apart from the scale bias between few-shot training data and diverse testing data, there are different kinds of agnostic bias. For instance, as reported in Table II, remote sensing scene methods such as the P-CNN [82], SAE-FSDet [6], FPD [83], SAGS [84], and MSOCL [85] can guarantee competitive results only under some certain settings rather than across all the settings. The reason for this is that although multioriented sample augmentation [82], multiscale instance feature extraction [85], gradual RPN design [6], and so on are proposed to achieve a significant improvement for FSOD task in the remote sensing domain, they can only solve some specific agnostic biases e.g., orientation bias, scale bias, and aspect ratio bias by training the detectors on limited available training images. Thus, only relying on few-shot training samples for model fine-tuning inevitably results in learning incomplete biases, which prevents the detector from working well for diverse testing scenarios, resulting in inferior generalizability. Moreover, although these specially designed modules have been shown to be effective for FSOD using few-shot training samples, feature representation learning is still unreliable for unseen testing objects, which makes it difficult to cope with agnostic bias. Therefore, introducing the universal domain-level knowledge from large-scale foundational models to improve the FSOD ability should be considered. To address this, we design controllable data generation and scalable feature representation learning based on a large-scale pretrained SD model to endow the detector with an imagination ability to prevent from learning incomplete agnostic bias from very limited training samples. The comparison of the SOTA detectors [6], [82], [85] with the proposed CGK-FSOD shows that the proposed CGK-FSOD achieves the best or second best performance in all the few-shot settings. This fully demonstrates that the conditional generative knowledge-guided VTP-DG and CMFE is beneficial for improving the FSOD performance. They are able to assist the detector in forming more generalized concepts for unseen object detection by relying solely on very limited training samples.

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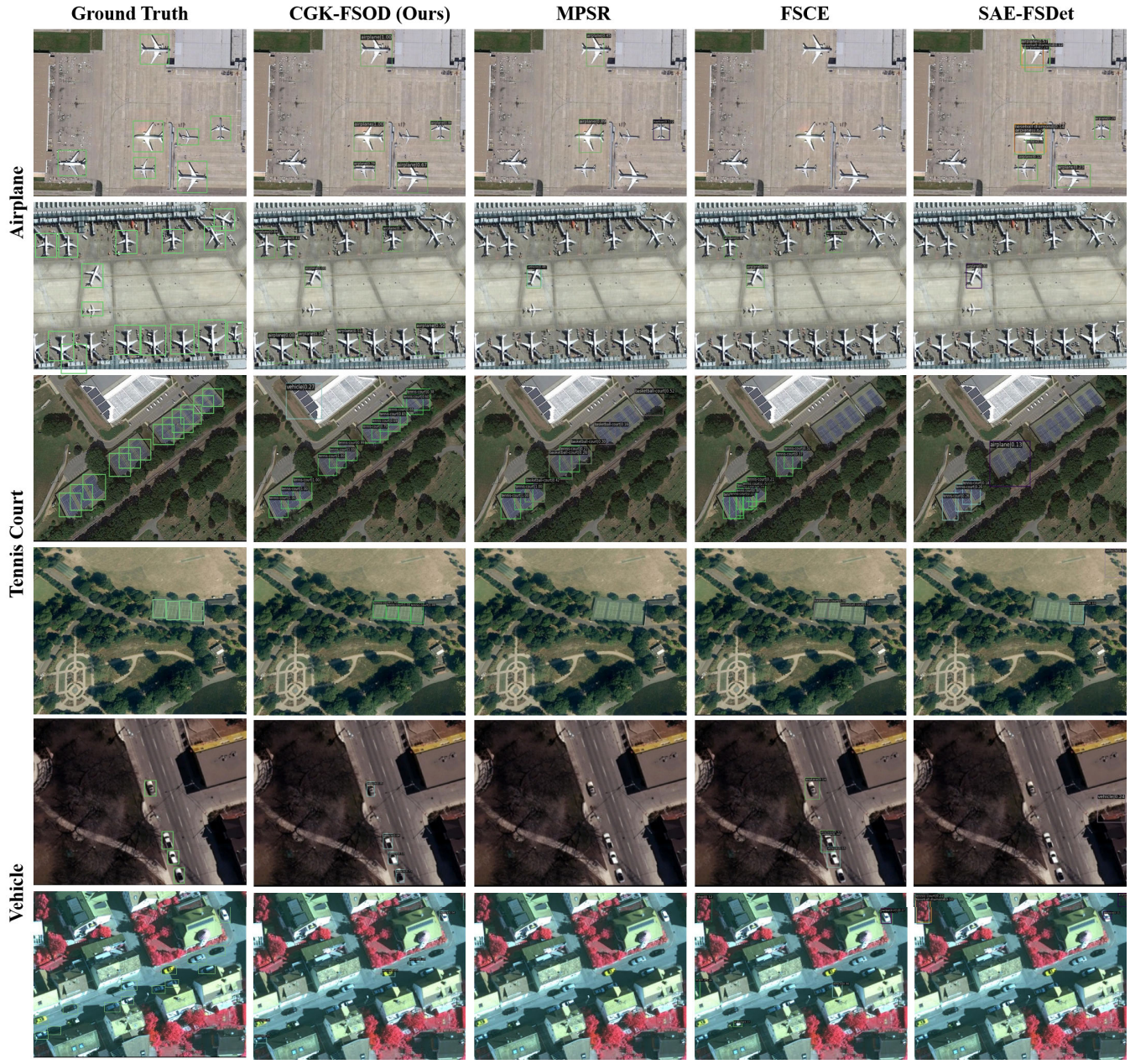


Fig. 4. Detection results from different FSOD methods under the setting of three-shot. Notably, the airplane, tennis court, and vehicle all belong to the set of novel categories in NV10 [13].

2) Results on NV10 Dataset: To illustrate the applicability of the proposed CGK-FSOD on different datasets, additional experiments are conducted on NV10 [13], and the detailed results are reported in Table III, aiming to observe the performance from different perspectives. Notably, the first column of Table III indicates the number of provided annotated samples (i.e., K-shot) for each new category, and the second column presents the detailed new categories and indicator of mAP. The 3rd–10th columns denote the reported results from different FSOD methods, including the AP of each novel category and the overall performance of mAP. Here, seven different FSOD methods, including TFA [33], FSCE [87], MPSR [34], Meta-RCNN [39], SAE-FSDet [6], FPD [83], and FsDetView [86], are selected for comparison.

Overall, the proposed CGK-FSOD has come out on top in terms of mAP across four different settings. From the view of a single category, the proposed CGK-FSOD can also provide the competitive AP compared with the other FSOD detectors. These results demonstrate the effectiveness of our proposed method for different datasets. In more detail, under the setting of three-shot learning, it can be found that our proposed CGK-FSOD has the highly advantageous, with an increase of approximately 20.3%–37.1% in mAP. Here, although the conventional FSOD detectors are designed to improve FSOD ability, the learnable space is still very limited to provide insufficient perception for diverse testing scenarios when the training samples are extremely scarce. As a comparison, our proposed CGK-FSOD can make use of

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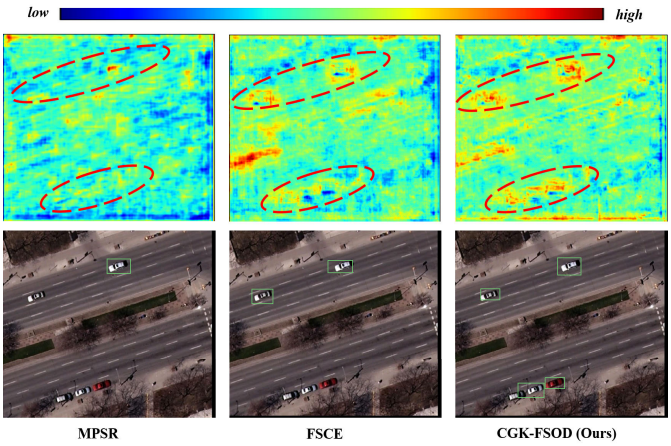


Fig. 5. Comparison on feature response of different FSOD detectors for the foreground objects of novel categories.

the generative pretrained SD model to derive more beneficial information according to the prior knowledge from limited training samples. Therefore, with an increasing number of K shots, conventional FSOD detectors, such as TFA [33], Meta-RCNN [39], and FsDetView [86], still have very limited performance improvement of FSOD. This illustrates that the few-shot detector designed for natural scenes cannot produce satisfactory results when directly applied for FSOD in the remote sensing domain. In contrast, the performance of FSCE [87], SAE-FSDet [6], and FPD [83] is gradually improved as the number of training samples increases. Then, this phenomenon indicates that when equipped with an advanced module design, the performance of FSOD is heavily influenced by the learnable data space which is related to the existing training samples. Meanwhile, it also proves that using the powerful generative ability to expand the learnable data space is very promising. In addition, to intuitively present the FSOD results of different detectors, several detection results for novel category detection in the setting of three-shot are shown in Fig. 4. By comparison, the proposed CGK-FSOD can detect more foreground objects than other detectors. However, a failure case can be found from the 3rd line of Fig. 4. The roof is mistakenly detected as a vehicle by our proposed CGK-FSOD. The reason is that the data expansion for the novel category vehicle impels the model to focus excessively on learning the texture diversity; thus, it could mislead the model to detect the regions that are very similar to the category of vehicle. To address this issue, a discussion about the balance of expanded data would be included in the ablation analysis. In addition to the intuitive detection results, we also visualized some feature maps that are from the neck network of different detectors, aiming to compare the feature response for foreground objects of new categories. As shown in the first line of Fig. 5, it can be found that the feature responses of MPSR [34] almost do not correctly focus on foreground objects, which explains why MPSR [34] misses detecting some vehicles as shown in the second line of Fig. 5. And then, although the feature map from FSCE [87] presents some high feature responses to foreground objects, the proposed CGK-FSOD more significantly responds to these vehicles. As a result, our proposed CGK-FSOD has the ability to detect more objects under the case of the fewest annotated training samples. Meanwhile, under the setting of 20 shots, the proposed CGK-FSOD also outperforms the SOTA remote sensing detector, i.e., SAE-FsDet [6] and achieves the best result, i.e., 48.3% mAP. This indicates that the proposed CGK-FSOD can provide a more stable and competitive FSOD performance regardless of the number of samples acquired.

C. Ablation Analysis

In this section, we further discuss the effects of the designed modules in the proposed CGK-FSOD. As a premise, a comprehensive analysis of the proposed modules is conducted on two datasets, i.e., NV10 [13] and DIOR† [12]. The results are reported in Table IV, where the 4th–7th columns denote the results (i.e., mAP of novel categories) of K-shot setting on NV10 [13], and the 8th–11th columns indicate the results (i.e., mAP of novel categories) of K-shot setting on DIOR† [12]. Particularly, to display the performance differences between our proposed CGK-FSOD and the baseline detector, we conduct multiround (i.e., five rounds) training and evaluate the result of each round. Then, mean and variance are calculated by these results, aiming to represent performance change in the form of "mean $\pm$ variance" in Table IV. First, as shown in the 3rd line, a baseline detector without any designed modules, i.e., MPSR [34] has the lowest performance across all the settings. Although MPSR [34] achieves good performance compared with the other detectors in Table II, its performance is still restricted by a fixed set of a few annotated samples. To address this limitation, the VTP-DG is first introduced into MPSR, which aims to establish a controllable data generation ability to enrich the learnable space of scarce training samples on the basis of the designed visual–textual prompts. According to the results in Table IV, the detector with the VTP-DG achieves a significant performance gain (i.e., 8.4%–22.1% mAP) across all the settings. These results indicate that realistic and controllable synthetic images from the designed VTP-DG have promising potential to assist the detector in learning the more complete agnostic bias information from very limited training samples for unseen object detection. Moreover, based on high-quality data generation, and aiming to improve the scalability for feature representation learning, a series of latent cross-modality generative features within the denoising generative process are reused by the designed CMFE, which can bring the visual–textual prompt-guided generative feature embedding into the feature fusion of the detector to enlarge the feature representation space. Subsequently, it can be incorporated with the designed VTP-DG to establish more generalized concepts for further improving FSOD performance. As shown in the last line of Table IV, the proposed CGK-FSOD can achieve the highest performance and outperform the baseline 19.7%, 16.8%, 12.3%, and 11.2% at four different settings of NV10 [13] and 24.2%, 15.8%, 17.3%, and 17.7% at four settings of DIOR† [12]. Moreover, to verify the applicability of the proposed models on different detectors, the proposed VTP-DG and CMFE are also implemented into two other FSOD detectors, i.e., meta-learning-based

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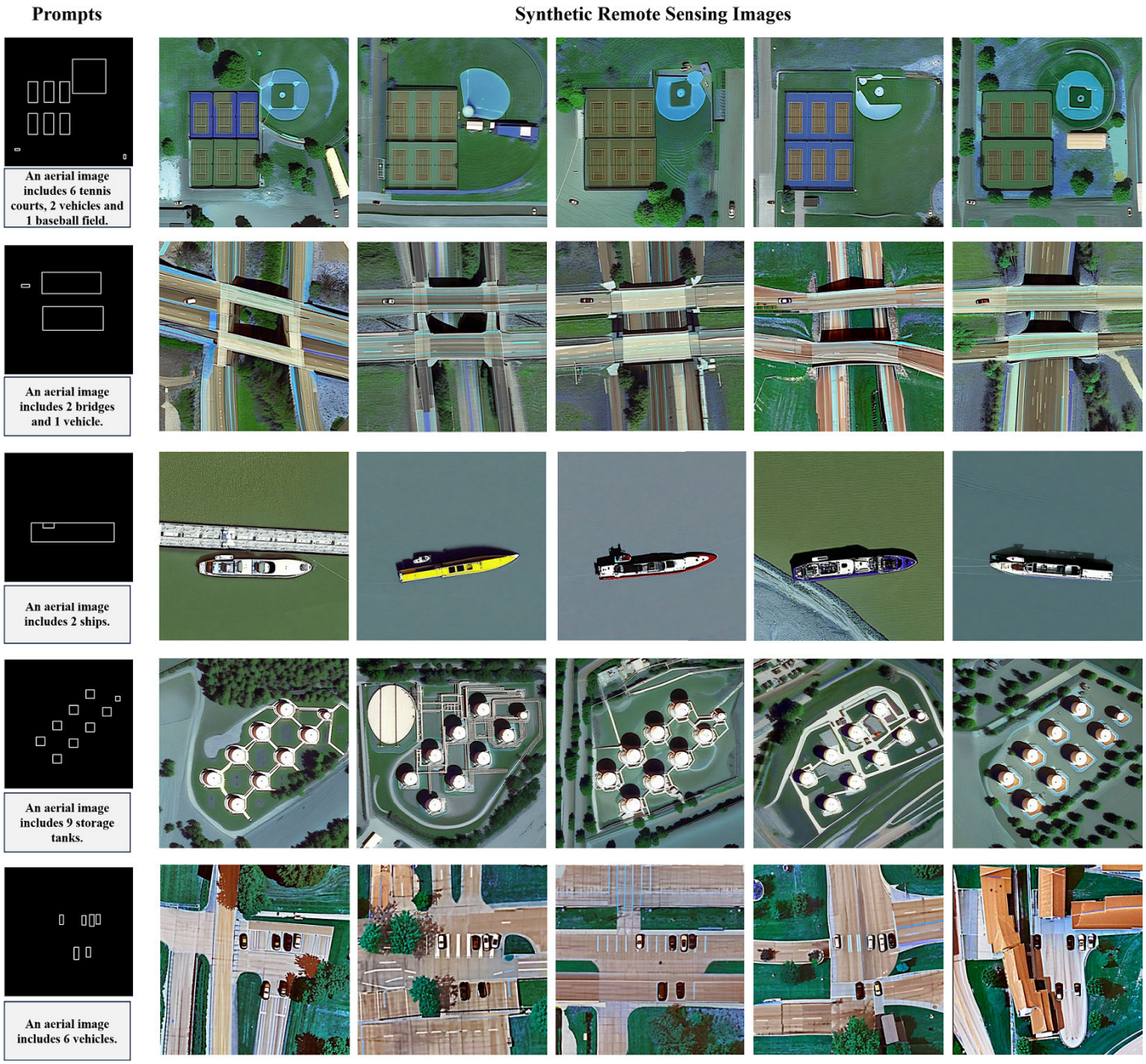


Fig. 6. Display of synthetic remote sensing images generated from VTP-DG by providing the corresponding visual–textual prompt.

TABLE IV

ABLATION EXPERIMENTS OF THE PROPOSED MODULES ON NV10 [13] AND DIOR† [12]. NOTABLY, †PRESENTS THAT SPLIT 1 OF DIOR [12] IS ADAPTED

Method	Modules		NV10				DIOR†			
	VTP-DG	CMFE	3-shot	5-shot	10-shot	20-shot	3-shot	5-shot	10-shot	20-shot
Baseline			20.7±1.56	24.8±0.89	32.2±0.10	37.1±1.01	17.4±0.07	26.2±0.37	27.4±0.35	29.8±0.24
	✓		39.2±1.61	39.8±1.62	40.6±1.15	45.8±2.61	39.5±1.01	41.5±0.30	44.4±0.76	46.3±0.22
	✓	✓	40.4±1.62	41.6±0.17	44.5±0.26	48.3±1.44	41.6±0.86	42.0±0.37	44.7±0.04	47.5±0.04

Meta-RCNN [39] and fine-tuning-based FSCE [87], whose structures are adjusted for adapting to the proposed VTP-DG and CMFE. As reported in Table V, it can be found that when equipped with the proposed VTP-DG and CMFE, Meta-RCNN [39] and FSCE [87] both achieve varying degrees of performance improvement, i.e., 0.9%–10.4% and 12.3%–29.4% mAP. Theoretically, the proposed VTP-DG and CMFE can play a role in injecting the controllable generative features from the pretrained SD model into the FSOD architecture for data expansion and scalable feature

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TABLE V

APPLICABILITY OF THE PROPOSED MODULES ON DIFFERENT DETECTORS, WHICH IS VERIFIED ON DIOR† [12]

Method	3-shot	5-shot	10-shot	20-shot
Meta-RCNN [39]	13.6±0.56	17.0±1.11	21.6±0.79	24.1±0.31
w. VTP-DG+CMFE	24.0±0.01	23.8±0.02	23.6±0.14	25.0±0.18
FSCE [87]	17.5±1.44	31.6±0.76	32.6±0.85	36.3±0.22
w. VTP-DG+CMFE	46.9±0.82	47.1±0.83	47.9±0.12	48.6±0.12

TABLE VI

COMPARISON OF DIFFERENT DATA EXPANSION METHODS ON NV10 [13] UNDER THE SETTINGS OF 3, 5, 10, AND 20 SHOTS

Baseline		✓	✓	✓	✓	✓
Aug.			✓			
VTP-DG	IS			✓	✓	✓
	DT				✓	✓
	FIF					✓
3-shot		20.7	16.6 (-4.1%)	34.7 (+14.0%)	36.3 (+15.6%)	39.2 (+18.5%)
5-shot		24.8	16.1 (-8.7%)	31.3 (+6.5%)	39.7 (+14.9%)	39.8 (+15.0%)
10-shot		32.2	23.0 (-9.2%)	29.3 (-2.9%)	37.5 (+5.3%)	40.6 (+8.4%)
20-shot		37.1	36.7 (-0.4%)	35.1 (-2.0%)	42.3 (+5.2%)	45.8 (+8.7%)

*Note: *Aug.*, IS, DT and FIF denotes the conventional data augmentation, image generation process by SD model, data transformation process and false instance filter, respectively.

TABLE VII

DISCUSSION ABOUT VISUAL AND TEXTUAL PROMPTS ON NV10 [13] UNDER THE SETTINGS OF 3, 5, 10, AND 20 SHOTS

Textual Prompt	w. class			✓
	w/o. class		✓	
Visual Prompt		✓	✓	✓
3-shot		38.9	39.5 (+0.6%)	40.4 (+1.5%)
5-shot		39.2	39.7 (+0.5%)	41.6 (+2.4%)
10-shot		41.4	42.1 (+0.7%)	44.5 (+3.1%)
20-shot		46.6	47.8 (+1.2%)	48.3 (+1.7%)

representation. Thereby, the proposed modules are effectively compatible with different FSOD detectors. Next, more detailed ablation studies about these proposed modules are provided in Sections IV-C1 and IV-C2.

1) Ablation Study on VTP-DG: The VTP-DG is designed to build controllable data generation ability, which can enrich the learnable space of scarce training samples to avoid overfitting while preventing the learning of incomplete agnostic bias from very limited training samples. To demonstrate this, first, as shown in Fig. 6, different types of synthetic optical remote sensing images are generated by the designed VTP-DG to present the generated quality. Through observation, the object positions and categories are controlled according to specific visual–textual prompts, whereas their background layouts and object appearances are flexible and diverse. These visualized examples indicate that the designed VTP-DG can provide sufficient high-quality synthetic object detection samples to support few-shot fine-tuning.

Second, apart from intuitive analysis, extra ablation experiments on comparing our designed VTP-DG and conventional data augmentation (namely *Aug.*) are carried out. Conventional data augmentation includes random scaling, rotating, shifting, and color jitter. To ensure fairness, the number of augmented samples remains consistent with the designed VTP-DG. The comparison results are listed in Table VI. As reported in the 2nd and 3rd columns, when the conventional data augmentation (*Aug.*) is applied for expanding few-shot fine-tuning samples, the performance has declined. The reason is that these augmented samples are restricted by the intrinsic background and object characteristics according to very limited training samples, which would make the detectors prone to learn the more intrinsic object appearances and contextual information, resulting in poor generalizability for diverse testing scenarios. Besides, regarding the proposed VTP-DG, a series of ablation experiments are conducted to progressively reveal the alterations in performance, as shown in the 4th–6th columns of Table VI, corresponding to three steps of VTP-DG including IS, DT and FIF. Among them, IS denotes the image generation process by SD model; DT denotes the data transformation process; and FIF represents the false instance filter. As reported in the 4th column of Table VI, the performance improvement attributed to IS is particularly pronounced under conditions of limited training samples, exhibiting a contrasting trend in comparison to conventional data augmentation. This demonstrates that the proposed VTP-DG can generate more effective object detection samples for adapting the agnostic bias from testing scenarios, even under the settings of three and five shots. As reported in the 5th column, with the addition of DT, one observation can be found that it can further bring the performance gain (i.e., 5.2%–15.6% mAP) across all the settings by enriching the distributions of object-level scale, orientation, and position. Subsequently, after introducing the FIF, it also can achieve a gradual performance improvement by filtering the low-quality generated instances. Overall, using the carefully designed steps, the proposed VTP-DG can provide a significant advantage in enriching the learnable space of scarce samples compared with conventional data augmentation.

Besides, at stage III of few-shot fine-tuning, only expanding the training samples for novel categories may lead to a sample imbalance issue between the base and novel categories, and it can make the detector prone to detecting the category of object that has more training samples. Hence, for further discussion, an ablation study is conducted, and the results are shown in Fig. 7. Wherein, the baseline represents the default setting of FSOD. The symbol of *w.* N represents the data expansion for novel categories; similarly, the symbol of *w.* $N + B$ denotes data expansion according to the novel and base categories. Through observing the curves of dashed blue and orange that indicate the base category performance, the mAP of the dashed orange curve presents an obvious decrease compared with the dashed blue curve. This phenomenon confirms that only expanding the training samples of novel categories would have a sample imbalance issue that impairs the performance of base categories. Furthermore, as shown in the dashed green curves that represent the simultaneous data expansion of novel and

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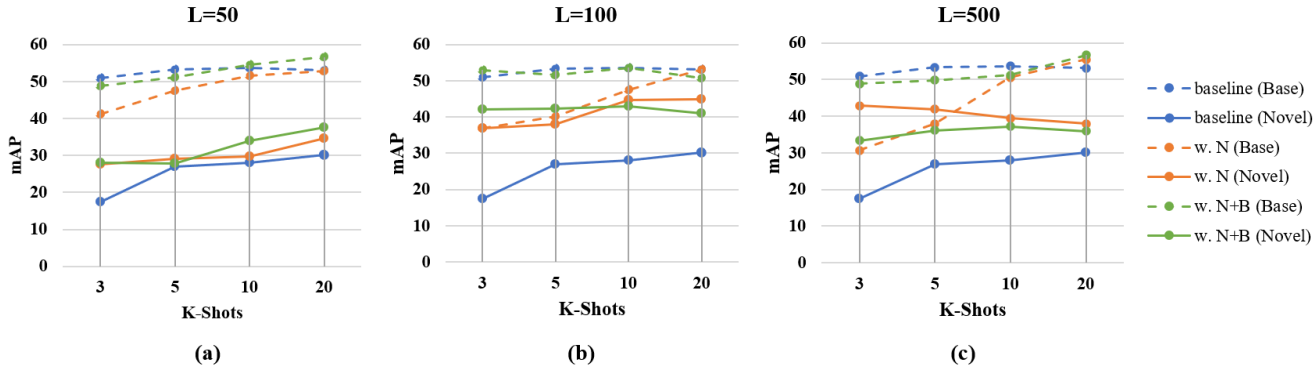


Fig. 7. Discussion about expanding the samples of base and novel categories by the designed VTP-DG. These dashed lines denote the mAP curves of base categories. The solid lines denote the mAP curves of novel categories. Notably, *w. N* denotes the data expansion for novel categories; *w.* $N + B$ denotes the data expansion according to the novel and base categories. (a) Number L of synthetic images is set to 50, (b) number L of synthetic images is set to 100, and (c) number L of synthetic images is set to 500.

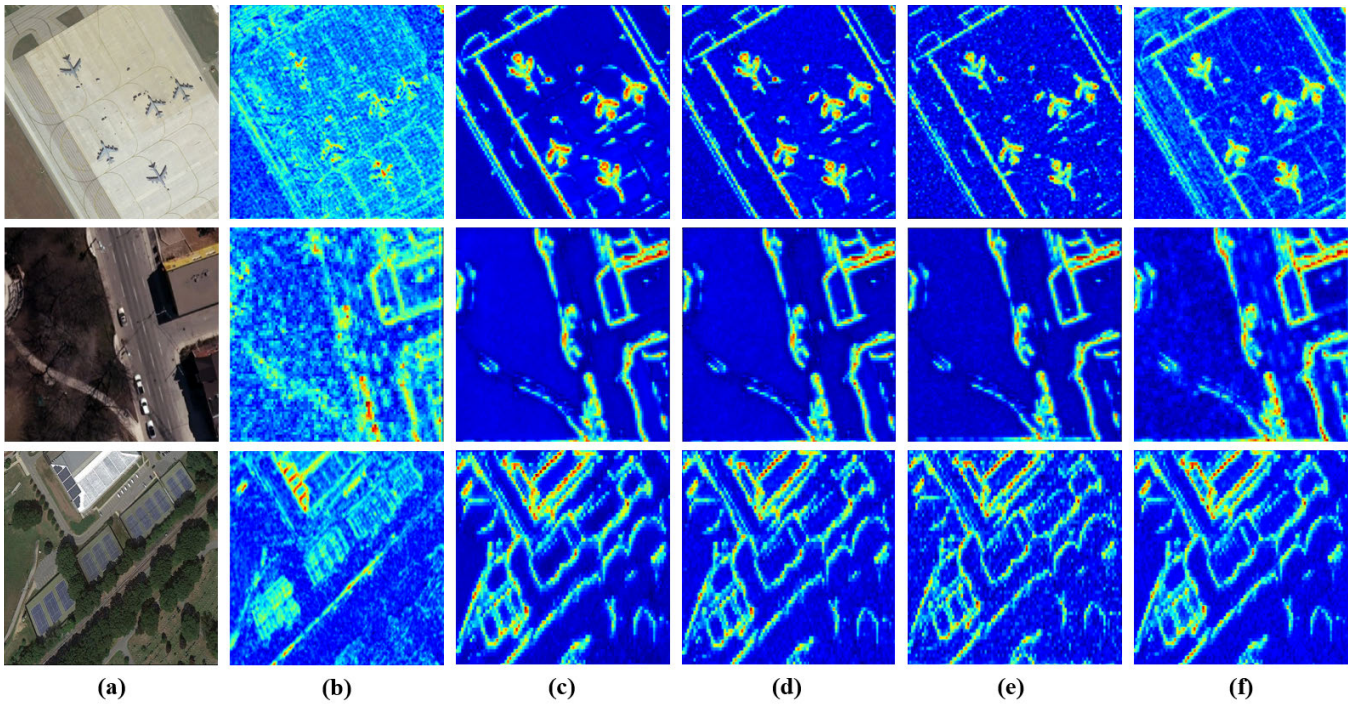


Fig. 8. Display of visualization results from the features of backbone network and controllable generative features of the SD model. (a) Original images. (b) Feature maps from the backbone network. (c) Feature maps that correspond to the time step of 1 from the SD model. (d) Feature maps that correspond to the time step of N/2 from the SD model. (e) Feature maps that correspond to the time step of N from the SD model. (f) Embedded cross-modality feature maps from CMFE.

base categories, the mAP of base categories can retain a similar performance with the dashed blue curves, which indicates that appropriate expansion of samples for base categories can enable the detector to maintain the detection ability for base categories. In view of novel categories, through observing the changes of solid curves, the solid blue curves all present the lowest mAP of novel categories. Along with the generation of synthetic images, the solid orange and green curves all present improvements. When expanding the samples of base and novel categories simultaneously, the mAP of novel categories has a slight decrease, which is compared with only expanding the training samples of novel categories. However, the slight mAP decrease in novel categories is in exchange for maintaining the detection capability of the base categories. Accordingly, this provides us with inspiration that if it is necessary to maintain the detection capability of base categories, more samples of base categories can be considered for incorporation during the few-shot fine-tuning phase.

Meanwhile, in Fig. 7, the number L of synthetic images is also selected as a variable for ablation analysis. Specifically, L is specified as 50, 100, and 500, respectively. As shown in Fig. 7(a), when L is set to 50, the performance gain from the proposed VTP-DG is very limited (i.e., 1.0%–10.2%) by comparing the solid blue curve with the solid orange and green curves. As L increases to 100, the performance appears a remarkable gain (i.e., 11.0%–25.0%), as shown in Fig. 7(b). Furthermore, when L increases to 500, by comparing with the results of $L = 100$, the performance gain has not been further improved, as shown in Fig. 7(c). According to this analysis, it can conclude that appropriate data expansion by the proposed VTP-DG can effectively promote the FSOD performance, but there exists a performance upper bound.

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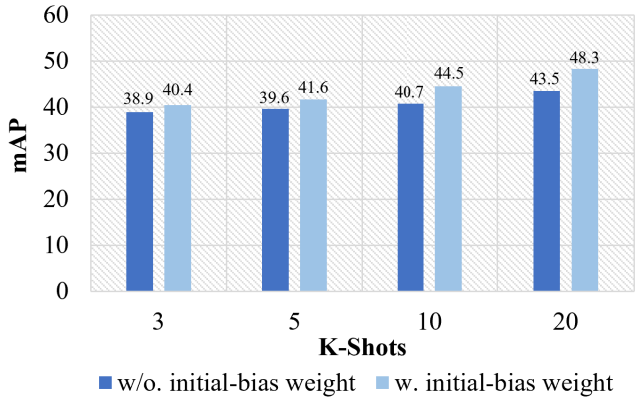


Fig. 9. Discussion about the effect of initial-bias weighting operation on the NV10 dataset [13].

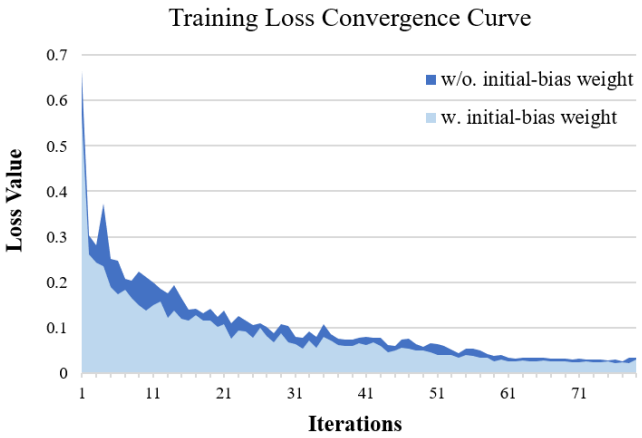


Fig. 10. Display of training loss curves under the setting of 20 shots. Notably, for a clearer display, the actual number of iterations is divided by 50 as the scale on the x-axis.

The reason for this is that the visual–textual prompts are provided by the knowledge of existing limited samples, thus it is still difficult to infinitely familiarize unknown testing scenarios.

2) Ablation Study on CMFE: In this section, more detailed discussions of the proposed CMFE are presented. First, to intuitively observe the effects of CMFE for scalable feature learning, some feature maps of detected images from NV10 [13] are visualized in Fig. 8. Here, Column (a) denotes original images of airplanes, vehicles, and tennis courts; Column (b) denotes the feature maps of F_1 from the backbone network; Columns (c)–(e) represent the feature maps of T_1^1, T_1^2, and T_1^3 from the SD model; Column (f) indicates the embedded cross-modality feature maps from CMFE. In comparison, the feature maps T_1^1, T_1^2, and T_1^3 from the SD model exhibit a better region localization ability than the feature map F_1 from the backbone network. The reason is that the visual prompt can play an important role in controlling valid data generation; thus, it also provides location-aware ability in the feature learning space. Moreover, from Column (c) to (e), as the number of time steps increased, these generative features within the denoising generative process present different scalabilities of feature representation; meanwhile, the location-aware ability is still maintained. When the aggregation of Columns (b)–(e) by the CMFE is finished, it can incorporate different scalabilities of controllable generative features into the feature fusion step of the detector and improve the FSOD performance. Here, it can be intuitively found from Column (f) that the foreground regions can be focused on to help localize the unseen objects. Moreover, the aggregated feature representation learning can be incorporated with controllable data generation at stage III of few-shot fine-tuning to form more generalized concepts for FSOD.

In addition, to demonstrate that textual-modality knowledge can help promote the perception of image content, the effects of textual prompts are validated by providing different kinds of textual prompts. Two types of descriptions are compared in Table VII. The first type, i.e., *w. class* indicates that the class names are filled in the textual template, and then adopting the image–text pair matching of Fig. 3 to obtain the potential textual prompts. The second type i.e., *w/o. class* represents the general textual template of "*An aerial image includes some objects.*" As shown in the 2nd, 3rd, and 4th columns of Table VII, with the addition of textual prompts, the performance of different FSOD settings achieves varying degrees of improvement. When the class name is further added, the gain from textual prompts is further increased, which demonstrates that textual-modality knowledge can positively affect the understanding of the image content and yield performance gain. Finally, to demonstrate the effect of the designed initial-bias weighting operation in the proposed CMFE, ablation analysis is executed to compare *w. initial-bias weight* and *w/o. initial-bias weight*, as shown in Fig. 9. By comparison, the light blue bars are consistently greater than the dark blue bars. This finding indicates that the CMFE with an initial-bias weight can facilitate FSOD performance during the same training period. As displayed in Fig. 10, to further study the impact of the initial-bias weighting operation, loss convergence curves are presented. One observation is that the initial value of the dark blue loss curve is larger than that of the light blue loss curve, which demonstrates that the initial-bias weighting operation can relieve the negative effect of feature differences between the SD model and the backbone of the detector. Moreover, by comparing the change trend of the loss curves, the availability of the initial-bias weighting operation leads to fast and stable convergence of the loss values. In conclusion, the designed initial-bias weighting operation is advantageous for embedding cross-modality generative features into the feature fusion step of the detector, which can increase the learning scalability for feature representation via few-shot fine-tuning.

V. Conclusion

In this article, a novel fine-tuning-based FSOD framework is proposed, in which an extra data generation stage is plugged into between base training and few-shot fine-tuning. Built upon the framework, the CGK-FSOD is proposed for unseen object detection from optical remote sensing imagery, which contains controllable data generation and scalable feature representation learning. Specifically, a novel VTP-DG is designed for

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controllable data generation by transferring pretrained models from natural scene into the remote sensing domain, which can enrich the learnable space of scarce training samples while avoiding the overfitting phenomenon and preventing the learning of incomplete agnostic bias from very limited training samples. Second, a new CMFE is designed for scalable feature representation learning by reusing a series of latent cross-modality generative features within the denoising generative process, which can introduce diverse generative features into the feature fusion step of the detector. Finally, extensive experiments are conducted on specific FSOD datasets, and the results indicate that the proposed CGK-FSOD can provide more stable and competitive FSOD performance. In general, effectively facilitating data generation for scarce training samples while endowing the detector with generative ability for feature representation learning is very promising for FSOD.

REFERENCES

[1] J. Wang, L. Zhu, Y. Yang, and L. Zhang, "Design, development and application of northeast Asia resources and environment scientific expedition data platform," *J. Resour. Ecol.*, vol. 2, no. 3, pp. 266–271, Jan. 2012.

[2] J. Wu, W. Xu, J. He, and M. Lan, "YOLO for penguin detection and counting based on remote sensing images," *Remote Sens.*, vol. 15, no. 10, p. 2598, May 2023.

[3] Z.-G. Wang, Q. Kang, Y.-J. Xun, Z.-Q. Shen, and C.-B. Cui, "Military reconnaissance application of high-resolution optical satellite remote sensing," *Proc. SPIE*, vol. 9299, pp. 301–305, Nov. 2014.

[4] Z. Zhao, P. Tang, L. Zhao, and Z. Zhang, "Few-shot object detection of remote sensing images via two-stage fine-tuning," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[5] N. Liu, X. Xu, T. Celik, Z. Gan, and H.-C. Li, "Transformation-invariant network for few-shot object detection in remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5625314.

[6] Y. Liu et al., "Few-shot object detection in remote-sensing images via label-consistent classifier and gradual regression," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5612114.

[7] Z. Zhou, L. Zhao, K. Ji, and G. Kuang, "A domain-adaptive few-shot SAR ship detection algorithm driven by the latent similarity between optical and SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5216318.

[8] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, vol. 139, 2021, pp. 8748–8763.

[9] A. Kirillov et al., "Segment anything," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2023, pp. 4015–4026.

[10] S. Liu et al., "Grounding DINO: Marrying DINO with grounded pre-training for open-set object detection," 2023, *arXiv:2303.05499*.

[11] L. Zhang and M. Agrawala, "Adding conditional control to text-to-image diffusion models," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2023, pp. 3836–3847.

[12] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, "Object detection in optical remote sensing images: A survey and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 296–307, Jan. 2020.

[13] G. Cheng, J. Han, P. Zhou, and L. Guo, "Multi-class geospatial object detection and geographic image classification based on collection of part detectors," *ISPRS J. Photogramm. Remote Sens.*, vol. 98, pp. 119–132, Dec. 2014.

[14] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1440–1448.

[15] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2961–2969.

[16] W. Li, Z. Chen, B. Li, D. Zhang, and Y. Yuan, "HTD: Heterogeneous task decoupling for two-stage object detection," *IEEE Trans. Image Process.*, vol. 30, pp. 9456–9469, 2021.

[17] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.

[18] A. Bochkovskiy, C.-Y. Wang, and H.-Y. Mark Liao, "YOLOv4: Optimal speed and accuracy of object detection," 2020, *arXiv:2004.10934*.

[19] Z. Ge, S. Liu, F. Wang, Z. Li, and J. Sun, "YOLOX: Exceeding YOLO series in 2021," 2021, *arXiv:2107.08430*.

[20] S. Liu and D. Huang, "Receptive field block net for accurate and fast object detection," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 385–400.

[21] X. Zhou, D. Wang, and P. Krähenbühl, "Objects as points," 2019, *arXiv:1904.07850*.

[22] G. Zhang, S. Lu, and W. Zhang, "CAD-Net: A context-aware detection network for objects in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 10015–10024, Dec. 2019.

[23] C. Li, G. Cheng, G. Wang, P. Zhou, and J. Han, "Instance-aware distillation for efficient object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5602011.

[24] T. Zhang et al., "Posterior instance injection detector for arbitrary-oriented object detection from optical remote-sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5623918.

[25] C. Zhang, J. Su, Y. Ju, K.-M. Lam, and Q. Wang, "Efficient inductive vision transformer for oriented object detection in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5616320.

[26] C. Xu et al., "Dynamic coarse-to-fine learning for oriented tiny object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 7318–7328.

[27] Z. Dong, M. Wang, Y. Wang, Y. Zhu, and Z. Zhang, "Object detection in high resolution remote sensing imagery based on convolutional neural networks with suitable object scale features," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 2104–2114, Mar. 2020.

[28] G. Cheng, Y. Si, H. Hong, X. Yao, and L. Guo, "Cross-scale feature fusion for object detection in optical remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 3, pp. 431–435, Mar. 2020.

[29] W. Zhang, L. Jiao, X. Liu, and J. Liu, "Multi-scale feature fusion network for object detection in VHR optical remote sensing images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2019, pp. 330–333.

[30] W. Jiang, Y. Sun, L. Lei, G. Kuang, and K. Ji, "Change detection of multisource remote sensing images: A review," *Int. J. Digit. Earth*, vol. 17, no. 1, Dec. 2024, Art. no. 2398051.

[31] H. Chen, Y. Wang, G. Wang, and Y. Qiao, "LSTD: A low-shot transfer detector for object detection," in *Proc. AAAI Conf. Artif. Intell.*, Apr. 2018, vol. 32, no. 1, pp. 2836–2843.

[32] F. Zhang, Y. Shi, Z. Xiong, and X. X. Zhu, "Few-shot object detection in remote sensing: Lifting the curse of incompletely annotated novel objects," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5603514.

[33] X. Wang, T. E. Huang, T. Darrell, J. E. Gonzalez, and F. Yu, "Frustratingly simple few-shot object detection," 2020, *arXiv:2003.06957*.

[34] J. Wu, S. Liu, D. Huang, and Y. Wang, "Multi-scale positive sample refinement for few-shot object detection," in *Proc. 16th Eur. Conf. Comput. Vis. (ECCV)*, Glasgow, U.K. Cham, Switzerland: Springer, Oct. 2020, pp. 456–472.

[35] A. Bar et al., "DETReg: Unsupervised pretraining with region priors for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 14585–14595.

[36] G. Han and S.-N. Lim, "Few-shot object detection with foundation models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 28608–28618.

[37] Y.-X. Wang, D. Ramanan, and M. Hebert, "Meta-learning to detect rare objects," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9925–9934.

[38] L. Wang, S. Mei, Y. Wang, J. Lian, Z. Han, and X. Chen, "Few-shot object detection with multilevel information interaction for optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5628014.

[39] X. Wu, D. Sahoo, and S. Hoi, "Meta-RCNN: Meta learning for few-shot object detection," in *Proc. 28th ACM Int. Conf. Multimedia*, Oct. 2020, pp. 1679–1687.

[40] M. Guo, Y. You, and F. Liu, "Discriminative prototype learning for few-shot object detection in remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5624313.

[41] G. Han, S. Huang, J. Ma, Y. He, and S.-F. Chang, "Meta Faster R-CNN: Towards accurate few-shot object detection with attentive feature alignment," in *Proc. AAAI Conf. Artif. Intell.*, 2022, pp. 780–789.

[42] B. Kang, Z. Liu, X. Wang, F. Yu, J. Feng, and T. Darrell, "Few-shot object detection via feature reweighting," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2019, pp. 8420–8429.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 23:09:43 UTC from IEEE Xplore. Restrictions apply.

[43] L. Karlinsky et al., "RepMet: Representative-based metric learning for classification and few-shot object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 5197–5206.

[44] Y. Yang, F. Wei, M. Shi, and G. Li, "Restoring negative information in few-shot object detection," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, Jan. 2020, pp. 3521–3532.

[45] M. Oquab et al., "DINOv2: Learning robust visual features without supervision," 2023, *arXiv:2304.07193*.

[46] H. Touvron et al., "LLaMA: Open and efficient foundation language models," 2023, *arXiv:2302.13971*.

[47] Y. Fu et al., "Cross-domain few-shot object detection via enhanced open-set object detector," in *Proc. Eur. Conf. Comput. Vis.*, Cham, Switzerland: Springer, Nov. 2024, pp. 247–264.

[48] L. Zhang, B. Zhang, B. Shi, J. Fan, and T. Chen, "Few-shot cross-domain object detection with instance-level prototype-based meta-learning," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 10, pp. 9078–9089, Oct. 2024.

[49] L. Girin, S. Leglaive, X. Bie, J. Diard, T. Hueber, and X. Alameda-Pineda, "Dynamical variational autoencoders: A comprehensive review," 2020, *arXiv:2008.12595*.

[50] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked autoencoders are scalable vision learners," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 16000–16009.

[51] D. Bank, N. Koenigstein, and R. Giryes, "Autoencoders," in *Machine Learning for Data Science Handbook: Data Mining and Knowledge Discovery Handbook*. Cham, Switzerland: Springer, 2023, pp. 353–374.

[52] K. Nguyen and S. Todorovic, "A self-supervised GAN for unsupervised few-shot object recognition," in *Proc. 25th Int. Conf. Pattern Recognit. (ICPR)*, Jan. 2021, pp. 3225–3231.

[53] E. Martinson, B. Furlong, and A. Gillies, "Training rare object detection in satellite imagery with synthetic GAN images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2021, pp. 2763–2770.

[54] Y. Zhang et al., "Inversion-based style transfer with diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 10146–10156.

[55] X. Bai and F. Xu, "Accelerating diffusion for SAR-to-optical image translation via adversarial consistency distillation," 2024, *arXiv:2407.06095*.

[56] T. Brooks, A. Holynski, and A. A. Efros, "InstructPix2Pix: Learning to follow image editing instructions," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 18392–18402.

[57] B. Kawar et al., "Imagic: Text-based real image editing with diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 6007–6017.

[58] J. Lu, X. Li, and K. Han, "RegionDrag: Fast region-based image editing with diffusion models," 2024, *arXiv:2407.18247*.

[59] B. Demirel, O. B. Baran, and R. G. Cinbis, "Meta-tuning loss functions and data augmentation for few-shot object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2023, pp. 7339–7349.

[60] J. Yao, T. Shi, X. Che, J. Yao, and L. Wu, "DA-FSOD: A novel data augmentation scheme for few-shot object detection," *IEEE Access*, vol. 11, pp. 92100–92110, 2023.

[61] S. Lin, K. Wang, X. Zeng, and R. Zhao, "An effective crop-paste pipeline for few-shot object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2023, pp. 4819–4827.

[62] J.-Y. Zhu, T. Park, P. Isola, and A. A. Efros, "Unpaired image-to-image translation using cycle-consistent adversarial networks," in *Proc. IEEE Int. Conf. Comput. Vis.*, Oct. 2017, pp. 2223–2232.

[63] R. San-Roman, E. Nachmani, and L. Wolf, "Noise estimation for generative diffusion models," 2021, *arXiv:2104.02600*.

[64] Z. Kong and W. Ping, "On fast sampling of diffusion probabilistic models," 2021, *arXiv:2106.00132*.

[65] J. Ho, C. Saharia, W. Chan, D. J. Fleet, M. Norouzi, and T. Salimans, "Cascaded diffusion models for high fidelity image generation," *J. Mach. Learn. Res.*, vol. 23, no. 47, pp. 1–33, 2022.

[66] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Proc. NIPS*, 2020, pp. 6840–6851.

[67] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 10684–10695.

[68] T. Yang, R. Wu, P. Ren, X. Xie, and L. Zhang, "Pixel-aware stable diffusion for realistic image super-resolution and personalized stylization," 2023, *arXiv:2308.14469*.

[69] A. Ramesh, P. Dhariwal, A. Nichol, C. Chu, and M. Chen, "Hierarchical text-conditional image generation with CLIP latents," 2022, *arXiv:2204.06125*.

[70] C. Schuhmann et al., "LAION-5B: An open large-scale dataset for training next generation image-text models," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 35, Jan. 2022, pp. 25278–25294.

[71] C. Mou et al., "T2I-adapter: Learning adapters to dig out more controllable ability for text-to-image diffusion models," in *Proc. AAAI Conf. Artif. Intell.*, Jan. 2023, pp. 4296–4304.

[72] R. Wang, Z. Chen, C. Chen, J. Ma, H. Lu, and X. S. Lin, "Compositional text-to-image synthesis with attention map control of diffusion models," in *Proc. AAAI Conf. Artif. Intell.*, Mar. 2024, vol. 38, no. 6, pp. 5544–5552.

[73] O. Avrahami et al., "SpaText: Spatio-textual representation for controllable image generation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 18370–18380.

[74] A. Voynov, K. Aberman, and D. Cohen-Or, "Sketch-guided text-to-image diffusion models," in *Proc. Special Interest Group Comput. Graph. Interact. Techn. Conf. Conf.*, Jul. 2023, pp. 1–11.

[75] D. Tang, X. Cao, X. Hou, Z. Jiang, J. Liu, and D. Meng, "CRS-diff: Controllable remote sensing image generation with diffusion model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5638714.

[76] S. Li et al., "ECNet: Effective controllable text-to-image diffusion models," 2024, *arXiv:2403.18417*.

[77] D. Li, J. Li, and S. C. H. Hoi, "BLIP-diffusion: Pre-trained subject representation for controllable text-to-image generation and editing," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 36, Jan. 2024, pp. 30146–30166.

[78] G.-S. Xia et al., "DOTA: A large-scale dataset for object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3974–3983.

[79] P. Vincent, H. Larochelle, Y. Bengio, and P.-A. Manzagol, "Extracting and composing robust features with denoising autoencoders," in *Proc. 25th Int. Conf. Mach. Learn. (ICML)*, 2008, pp. 1096–1103.

[80] W. Zhao, Y. Rao, Z. Liu, B. Liu, J. Zhou, and J. Lu, "Unleashing text-to-image diffusion models for visual perception," in *Proc. Int. Conf. Comput. Vis.*, 2023, pp. 5729–5739.

[81] S. Xie and Z. Tu, "Holistically-nested edge detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1395–1403.

[82] G. Cheng et al., "Prototype-CNN for few-shot object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–10, 2021.

[83] Z. Wang, B. Yang, H. Yue, and Z. Ma, "Fine-grained prototypes distillation for few-shot object detection," in *Proc. AAAI Conf. Artif. Intell.*, Mar. 2024, vol. 38, no. 6, pp. 5859–5866.

[84] Y. Zhang, B. Zhang, and B. Wang, "Few-shot object detection with self-adaptive global similarity and two-way foreground stimulator in remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 7263–7276, 2022.

[85] J. Chen, D. Qin, D. Hou, J. Zhang, M. Deng, and G. Sun, "Multiscale object contrastive learning-derived few-shot object detection in VHR imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5635615.

[86] Y. Xiao, V. Lepetit, and R. Marlet, "Few-shot object detection and viewpoint estimation for objects in the wild," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 3, pp. 3090–3106, Mar. 2022.

[87] B. Sun, B. Li, S. Cai, Y. Yuan, and C. Zhang, "FSCE: Few-shot object detection via contrastive proposal encoding," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 7352–7362.

[88] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[89] F. Liu et al., "RemoteCLIP: A vision language foundation model for remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5622216.

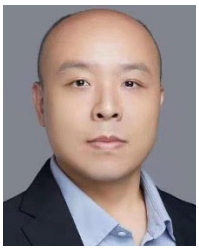
[90] M. Everingham, S. M. A. Eslami, L. Van Gool, C. K. I. Williams, J. Winn, and A. Zisserman, "The PASCAL visual object classes challenge: A retrospective," *Int. J. Comput. Vis.*, vol. 111, no. 1, pp. 98–136, Jan. 2015.

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